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ENERGY CONVERSION-I

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LECTURE-08

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Inspiring Excellence

Stephen J. Chapman
5th Edition
Chapter 6
Induction Motors

Chapter 6: Induction Motors

- Construction of Induction Motors
- Basic Induction Motor Concepts
- Equivalent circuit of an induction motor
- Power and Torque in induction motor
- **Torque-speed characteristics**
- Starting Induction Motors
- Speed Control of Induction Motors

Induced Torque from a Physical Standpoint

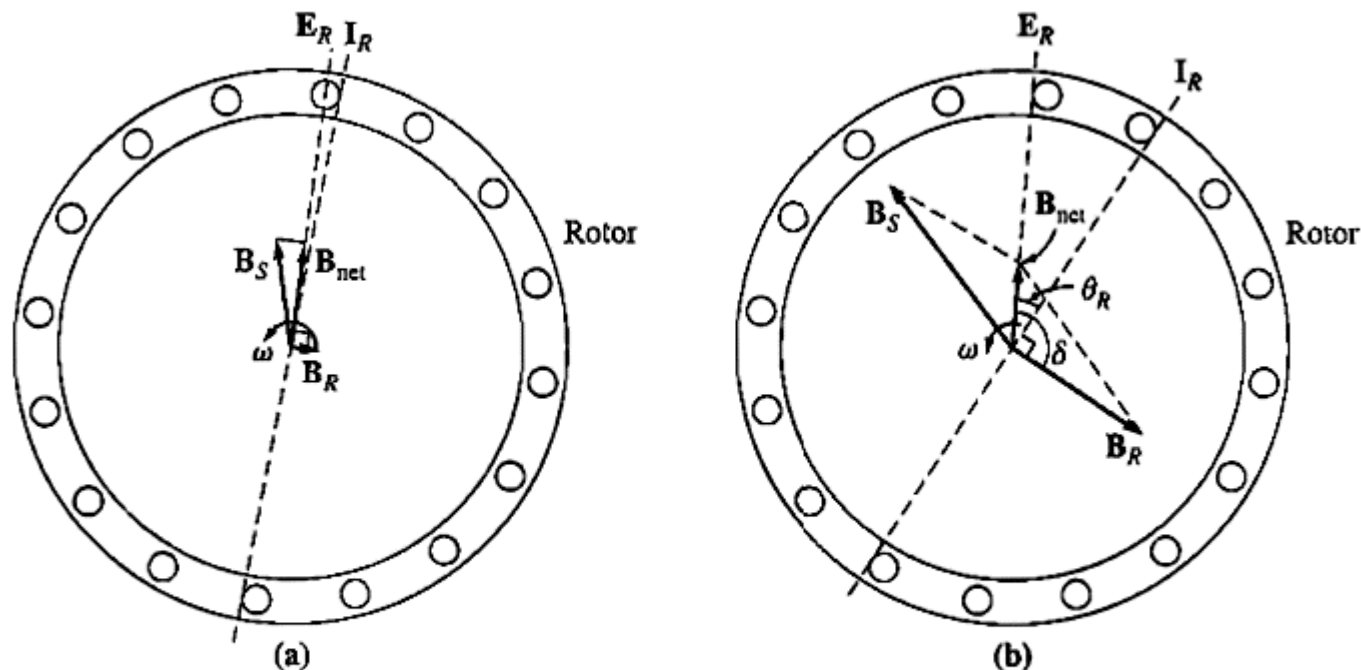


FIGURE 6-15

(a) The magnetic fields in an induction motor under light loads. (b) The magnetic fields in an induction motor under heavy loads.

Figure 6-15a shows a cage rotor induction motor that is initially operating at no load and therefore very nearly at synchronous speed. The net magnetic field $\mathbf{B}_{\text{net}}$ in this machine is produced by the magnetization current $\mathbf{I}_M$ flowing in the motor's equivalent circuit (see Figure 6-12). The magnitude of the magnetization current and hence of $\mathbf{B}_{\text{net}}$ is directly proportional to the voltage $\mathbf{E}_1$. If $\mathbf{E}_1$ is constant, then the net magnetic field in the motor is constant. In an actual machine, $\mathbf{E}_1$ varies as the load changes, because the stator impedances R_1 and X_1 cause varying voltage drops with varying load. However, these drops in the stator windings are relatively small, so $\mathbf{E}_1$ (and hence $\mathbf{I}_M$ and $\mathbf{B}_{\text{net}}$) is approximately constant with changes in load.

Figure 6-15a shows the induction motor at no load. At no load, the rotor slip is very small, and so the relative motion between the rotor and the magnetic fields is very small and the rotor frequency is also very small. Since the relative motion is small, the voltage $\mathbf{E}_R$ induced in the bars of the rotor is very small, and the resulting current flow $\mathbf{I}_R$ is small. Also, because the rotor frequency is so very small, the reactance of the rotor is nearly zero, and the maximum rotor current $\mathbf{I}_R$ is almost in phase with the rotor voltage $\mathbf{E}_R$. The rotor current thus produces a small magnetic field $\mathbf{B}_R$ at an angle just slightly greater than 90° behind the net magnetic field $\mathbf{B}_{\text{net}}$. Notice that the stator current must be quite large even at no load, since it must supply most of $\mathbf{B}_{\text{net}}$. (This is why induction motors have large

no-load currents compared to other types of machines. The no-load current of an induction motor is usually 30–60 percent of the full-load current.)

The induced torque, which keeps the rotor turning, is given by the equation

$$\tau_{\text{ind}} = k \mathbf{B}_R \times \mathbf{B}_{\text{net}} \quad (3-60)$$

Its magnitude is given by

$$\tau_{\text{ind}} = k \mathbf{B}_R \mathbf{B}_{\text{net}} \sin \delta \quad (3-61)$$

Since the rotor magnetic field is very small, the induced torque is also quite small—just large enough to overcome the motor's rotational losses.

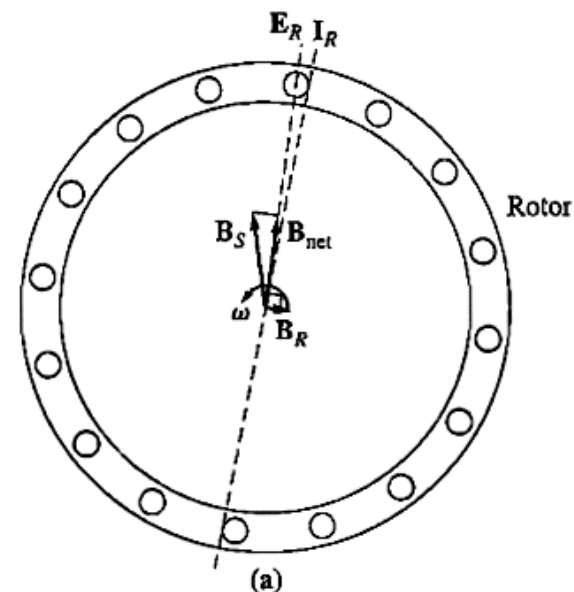


FIGURE 6--15
(a) The magnetic fields in an induction motor under light loads

Now suppose the induction motor is loaded down (Figure 6–15b). As the motor's load increases, its slip increases, and the rotor speed falls. Since the rotor speed is slower, there is now *more relative motion* between the rotor and the stator magnetic fields in the machine. Greater relative motion produces a stronger rotor voltage E_R which in turn produces a larger rotor current I_R . With a larger rotor current, the rotor magnetic field B_R also increases. However, the angle of the rotor current and B_R changes as well. Since the rotor slip is larger, the rotor frequency rises ($f_{re} = sf_{se}$), and the rotor's reactance increases ($\omega_{re}L_R$). Therefore, the rotor current now lags further behind the rotor voltage, and the rotor magnetic field shifts with the current. Figure 6–15b shows the induction motor operating at a fairly high load. Notice that the rotor current has increased and that the angle δ has increased. The increase in B_R tends to increase the torque, while the increase in angle δ tends to decrease the torque (τ_{ind} is proportional to $\sin \delta$, and $\delta > 90^\circ$). Since the first effect is larger than the second one, the overall induced torque increases to supply the motor's increased load.

When does an induction motor reach pullout torque? This happens when the point is reached where, as the load on the shaft is increased, the $\sin \delta$ term decreases more than the B_R term increases. At that point, a further increase in load decreases τ_{ind} , and the motor stops.

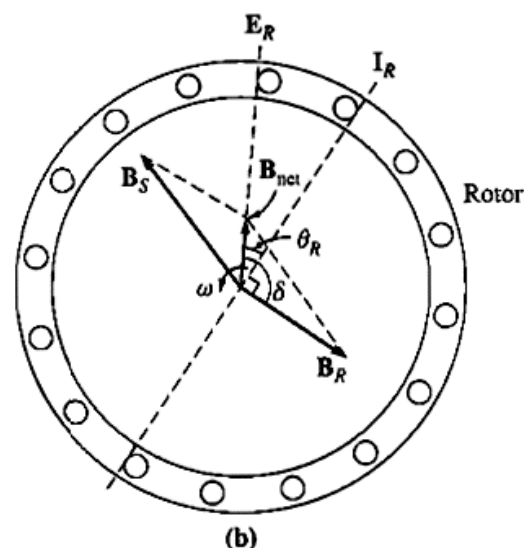


FIGURE 6—15 (b)
The magnetic
fields in an
induction motor
under heavy loads.

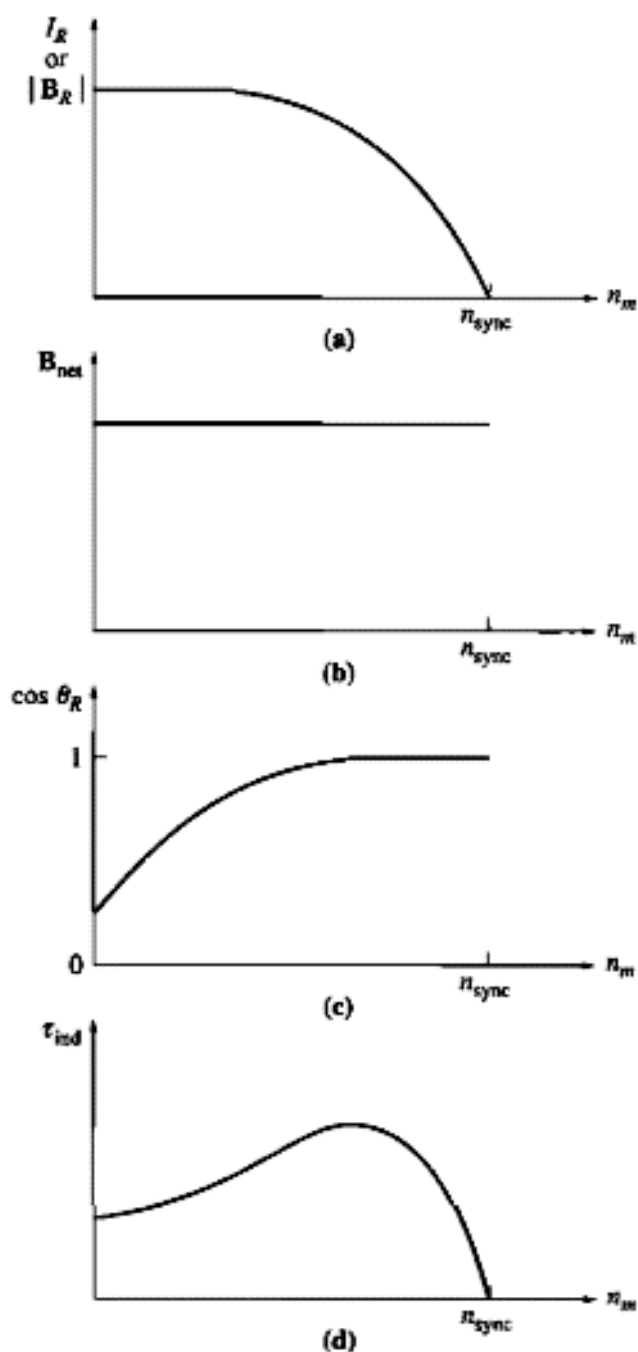


FIGURE 6-16

Graphical development of an induction motor torque- speed characteristic.

(a) Plot of rotor current (and thus $|B_R|$) versus speed for an induction motor; (b) plot of net magnetic field versus speed for the motor; (c) plot of rotor power factor versus speed for the motor; (d) the resulting torque-speed characteristic.

Remember that the magnitude of the induced torque in the machine is given by

$$\tau_{\text{ind}} = k B_R B_{\text{net}} \sin \delta \quad (3-61)$$

Each term in this expression can be considered separately to derive the overall machine behavior. The individual terms are

1. B_R . The rotor magnetic field is directly proportional to the current flowing in the rotor, as long as the rotor is unsaturated. The current flow in the rotor increases with increasing slip (decreasing speed) according to Equation (6-13). This current flow was plotted in Figure 6-11 and is shown again in Figure 6-16a.
2. B_{net} . The net magnetic field in the motor is proportional to E_1 and therefore is approximately constant (E_1 actually decreases with increasing current flow, but this effect is small compared to the other two, and it will be ignored in this graphical development). The curve for B_{net} versus speed is shown in Figure 6-16b.
3. $\sin \delta$. The angle δ between the net and rotor magnetic fields can be expressed in a very useful way. Look at Figure 6-15b. In this figure, it is clear that *the angle δ is just equal to the power-factor angle of the rotor plus 90°* :

$$\delta = \theta_R + 90^\circ \quad (6-38)$$

Therefore, $\sin \delta = \sin (\theta_R + 90^\circ) = \cos \theta_R$. This term is the power factor of the rotor. The rotor power-factor angle can be calculated from the equation

$$\theta_R = \tan^{-1} \frac{X_R}{R_R} = \tan^{-1} \frac{sX_{R0}}{R_R} \quad (6-39)$$

The resulting rotor power factor is given by

$$\text{PF}_R = \cos \theta_R$$

$$\text{PF}_R = \cos \left(\tan^{-1} \frac{sX_{R0}}{R_R} \right) \quad (6-40)$$

characteristic of an induction motor derived in this fashion is shown in **Figure 6–16d**.

This characteristic curve can be divided roughly into three regions. The first region is the **low-slip region** of the curve. In the low-slip region, the motor slip increases approximately linearly with increased load, and the rotor mechanical speed decreases approximately linearly with load. In this region of operation, the rotor reactance is negligible, so the rotor power factor is approximately unity, while the rotor current increases linearly with slip. *The entire normal steady-state operating range of an induction motor is included in this linear low-slip region.* Thus in normal operation, an induction motor has a linear speed droop.

The second region on the induction motor's curve can be called the *moderate-slip region*. In the **moderate-slip region**, the rotor frequency is higher than before, and the rotor reactance is on the same order of magnitude as the rotor resistance. In this region, the rotor current no longer increases as rapidly as before, and the power factor starts to drop. The peak torque (the *pullout torque*) of the motor occurs at the point where, for an incremental increase in load, the increase in the rotor current is exactly balanced by the decrease in the rotor power factor.

The third region on the induction motor's curve is called the *high-slip region*. In the **high-slip region**, the induced torque actually decreases with increased load, since the increase in rotor current is completely overshadowed by the decrease in rotor power factor.

For a typical induction motor, the pullout torque on the curve will be 200 to 250 percent of the rated full-load torque of the machine, and the *starting torque* (the torque at zero speed) will be 150 percent or so of the full-load torque. Unlike a synchronous motor, the induction motor can start with a full load attached to its shaft.

Torque, power and Thevenin's Theorem

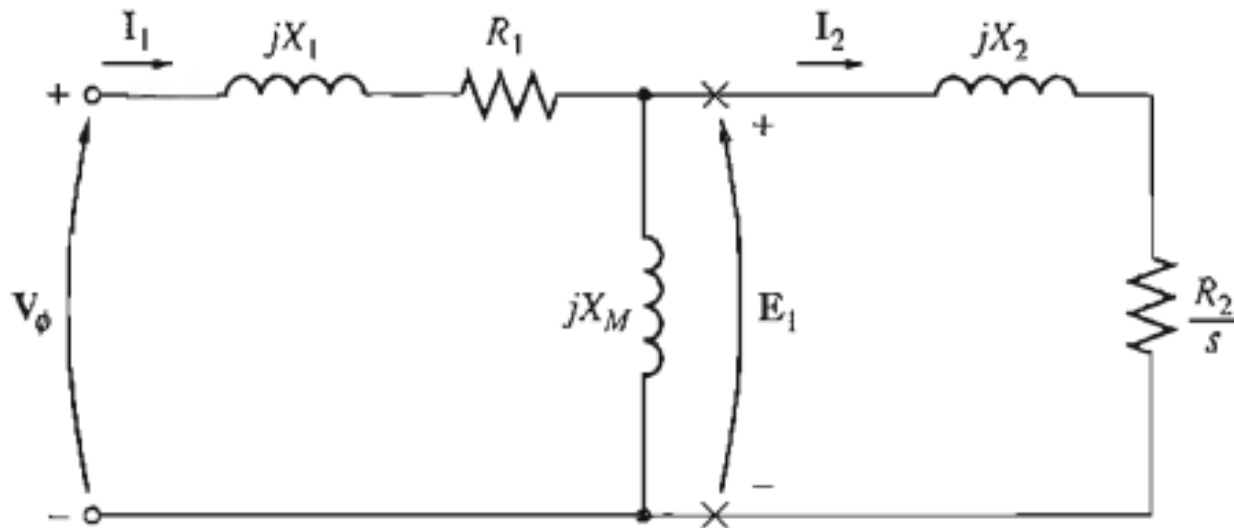
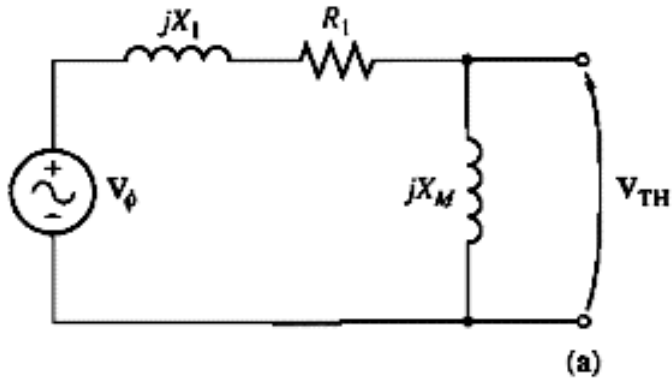


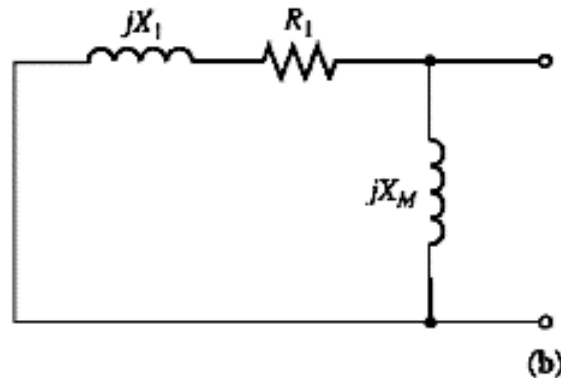
FIGURE 6-17

Per-phase equivalent circuit of an induction motor.



$$V_{TH} = \frac{jX_M}{R_1 + jX_1 + jX_M} V_\phi$$

$$V_{TH} = \frac{X_M}{\sqrt{R_1^2 + (X_1 + X_M)^2}} V_\phi$$



$$Z_{TH} = \frac{jX_M (R_1 + jX_1)}{R_1 + j(X_1 + X_M)}$$

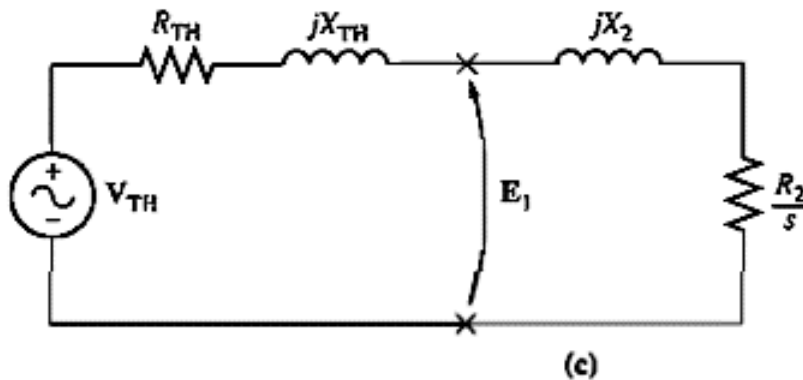


FIGURE 6-18

(a) The Thevenin equivalent voltage of an induction motor input circuit. (b) The Thevenin equivalent impedance of the input circuit. (c) The resulting simplified equivalent circuit of an induction motor.

Rotor-induced torque is given by:

$$\tau_{\text{ind}} = \frac{3V_{\text{TH}}^2 R_2/s}{\omega_{\text{sync}}[(R_{\text{TH}} + R_2/s)^2 + (X_{\text{TH}} + X_2)^2]} \quad (6-50)$$

Torque-speed characteristics

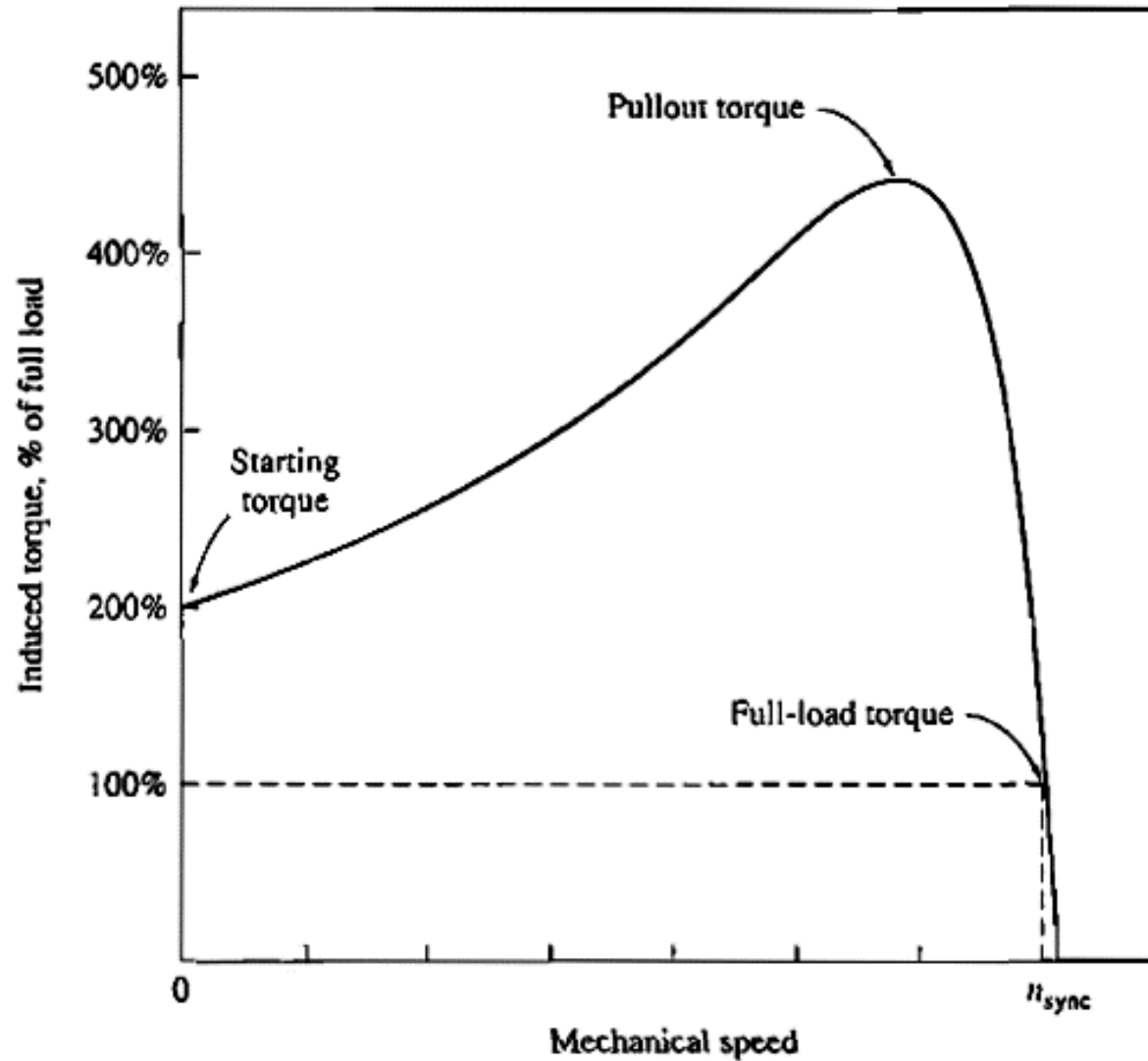


FIGURE 6-19
A typical induction motor torque-speed characteristic curve.

Comments

1. The induced torque is **zero** at **synchronous speed**.
2. The curve is **nearly linear** between **no-load** and **full load**. In this range, the rotor resistance is much greater than the reactance, so the rotor current, torque increase linearly with the slip.
3. There is a **maximum possible torque** that can't be exceeded. This torque is called **pullout torque** and is **2 to 3 times** the **rated full-load torque**.

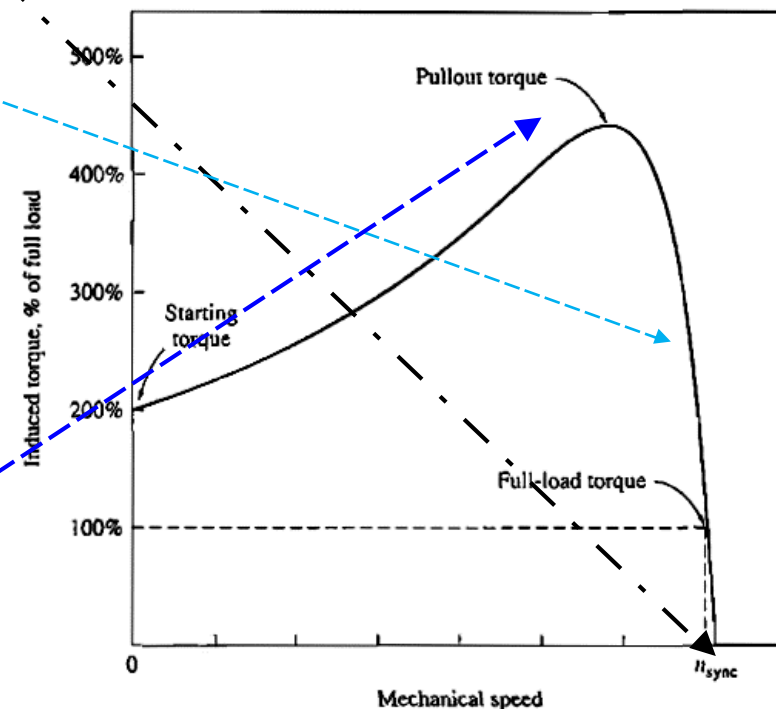
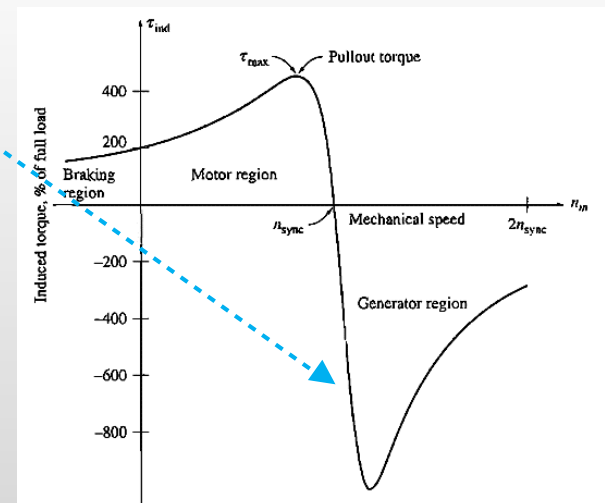
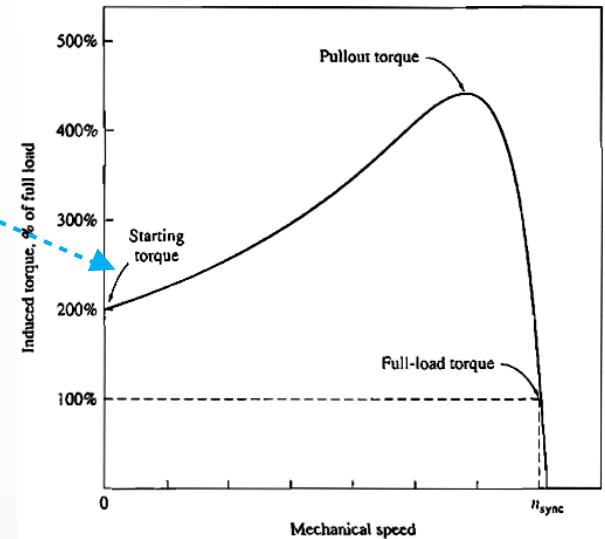


FIGURE 6-19

A typical induction motor torque-speed characteristic curve.

Comments

4. The starting torque of the motor is slightly higher than its full-load torque, so the motor will start carrying any load it can supply at full load.
5. The torque of the motor for a given slip varies as the square of the applied voltage.
6. If the rotor is driven faster than synchronous speed it will run as a generator, converting mechanical power to electric power.
7. If the motor is turning backward relative to the direction of the magnetic fields, the induced torque in the machine will stop the machine very rapidly and will try to rotate it in the other direction. Since reversing the direction of magnetic field rotation is simply a matter of switching any two stator phases, this fact can be used as a way to very rapidly stop an induction motor. The act of switching two phases in order to stop the motor very rapidly is called *plugging*.



Complete Curve

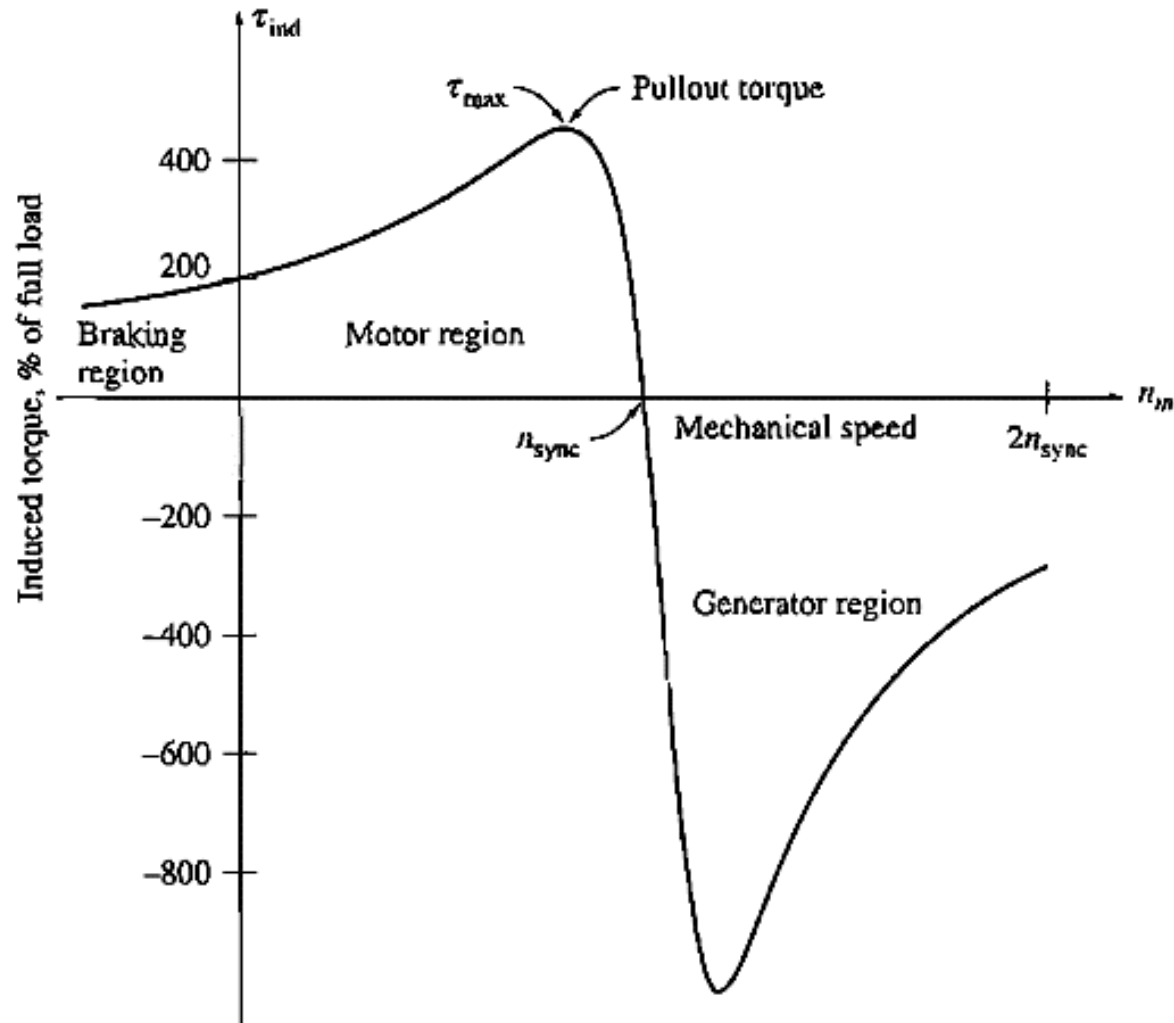


FIGURE 6-20

Induction motor torque-speed characteristic curve, showing the extended operating ranges (braking region and generator region).

Maximum (Pullout) torque

- Since the air-gap power is equal to the power consumed in the resistor R_2/s , the *maximum induced torque will occur when the power consumed by that resistor is maximum.*
- This condition occurs when R_2/s equals the magnitude of the impedance $R_{TH} + j(X_{TH} + X_2)$

The equivalent source impedance in the circuit is

$$Z_{\text{source}} = R_{TH} + jX_{TH} + jX_2 \quad (6-51)$$

the slip at pullout torque is given by

$$\frac{R_2}{s} = \sqrt{R_{TH}^2 + (X_{TH} + X_2)^2} \quad (6-52)$$

the slip at pullout torque is given by

$$s_{\text{max}} = \frac{R_2}{\sqrt{R_{TH}^2 + (X_{TH} + X_2)^2}} \quad (6-53)$$

Maximum (Pullout) torque

The corresponding maximum torque of an induction motor equals

$$\tau_{\max} = \frac{3V_{TH}^2}{2\omega_{\text{sync}}[R_{TH} + \sqrt{R_{TH}^2 + (X_{TH} + X_2)^2}]} \quad (6-54)$$

The slip at maximum torque is directly proportional to the rotor resistance R_2

The maximum torque is independent of R_2

Rotor resistance can be increased by inserting external resistance in the rotor of a **wound-rotor** induction motor.

The
value of the maximum torque remains unaffected
but
the speed at which it occurs can be controlled.

Example 6.4

A two-pole, 50-Hz induction motor supplies 15kW to a load at a speed of 2950 rpm.

- 1. What is the motor's slip?*
- 2. What is the induced torque in the motor in N.m under these conditions?*
- 3. What will be the operating speed of the motor if its torque is doubled?*
- 4. How much power will be supplied by the motor when the torque is doubled?*

Solution

$$1. \quad n_{sync} = \frac{120 f_e}{P} = \frac{120 \times 50}{2} = 3000 \text{ rpm}$$

$$s = \frac{n_{sync} - n_m}{n_{sync}} = \frac{3000 - 2950}{3000} = 0.0167 \text{ or } 1.67\%$$

2. $\because$ no P_{f+W} given

$\therefore$ assume $P_{conv} = P_{load}$ and $\tau_{ind} = \tau_{load}$

$$\tau_{ind} = \frac{P_{conv}}{\omega_m} = \frac{15 \times 10^3}{2950 \times \frac{2\pi}{60}} = 48.6 \text{ N.m}$$

Solution

3. *In the low-slip region, the torque-speed curve is linear and the induced torque is direct proportional to slip. So, if the torque is doubled the new slip will be 3.33% and the motor speed will be*

$$n_m = (1 - s)n_{sync} = (1 - 0.0333) \times 3000 = 2900 \text{ rpm}$$

4. $P_{conv} = \tau_{ind} \omega_m$

$$= (2 \times 48.6) \times (2900 \times \frac{2\pi}{60}) = 29.5 \text{ kW}$$